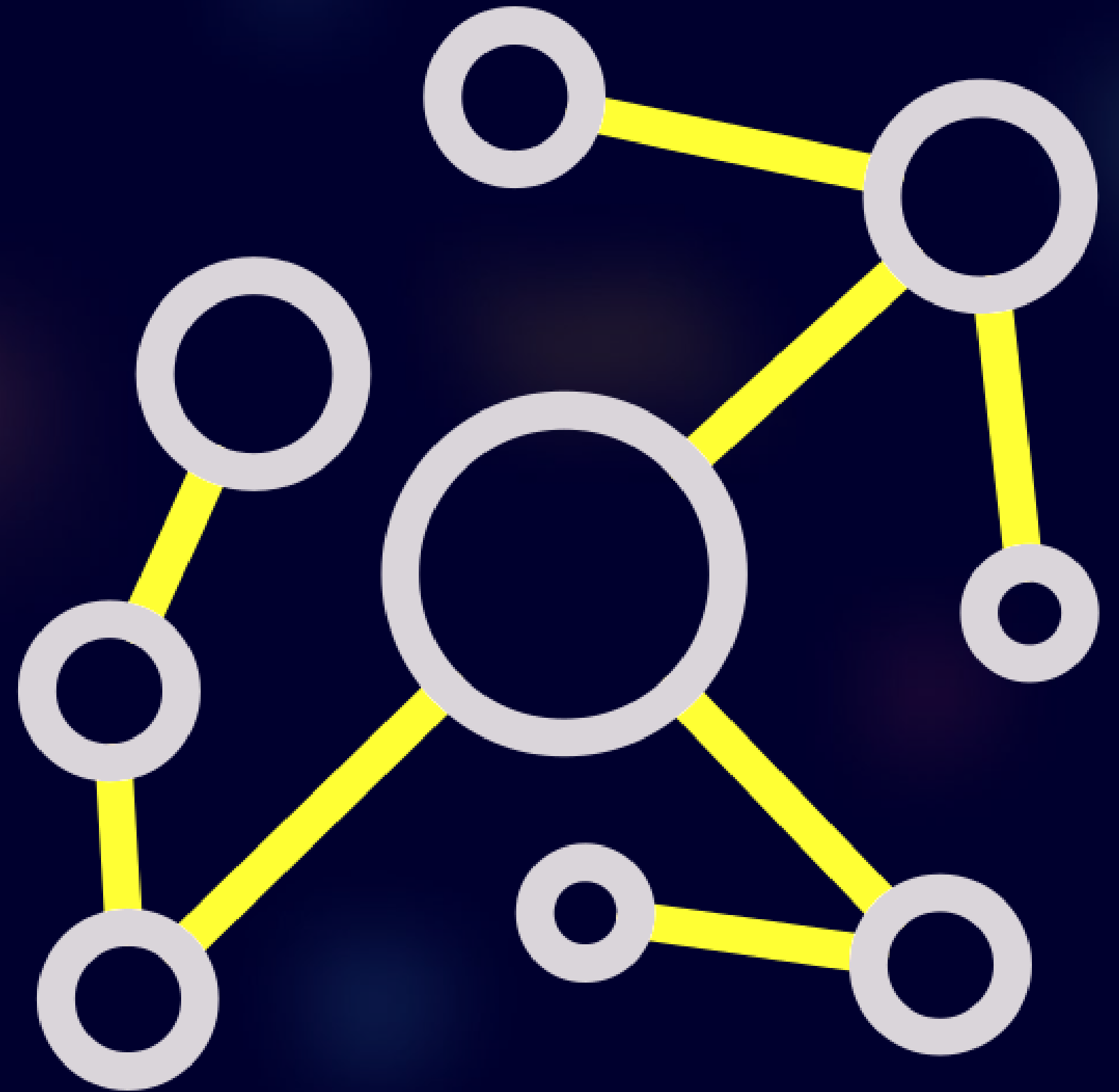


Developing Open Interoperable Semantic Resources

Grupo 6 · Santiago Rodríguez · Daniel Martín ·
Daniel Jiménez · Alejandro de Hermenegildo



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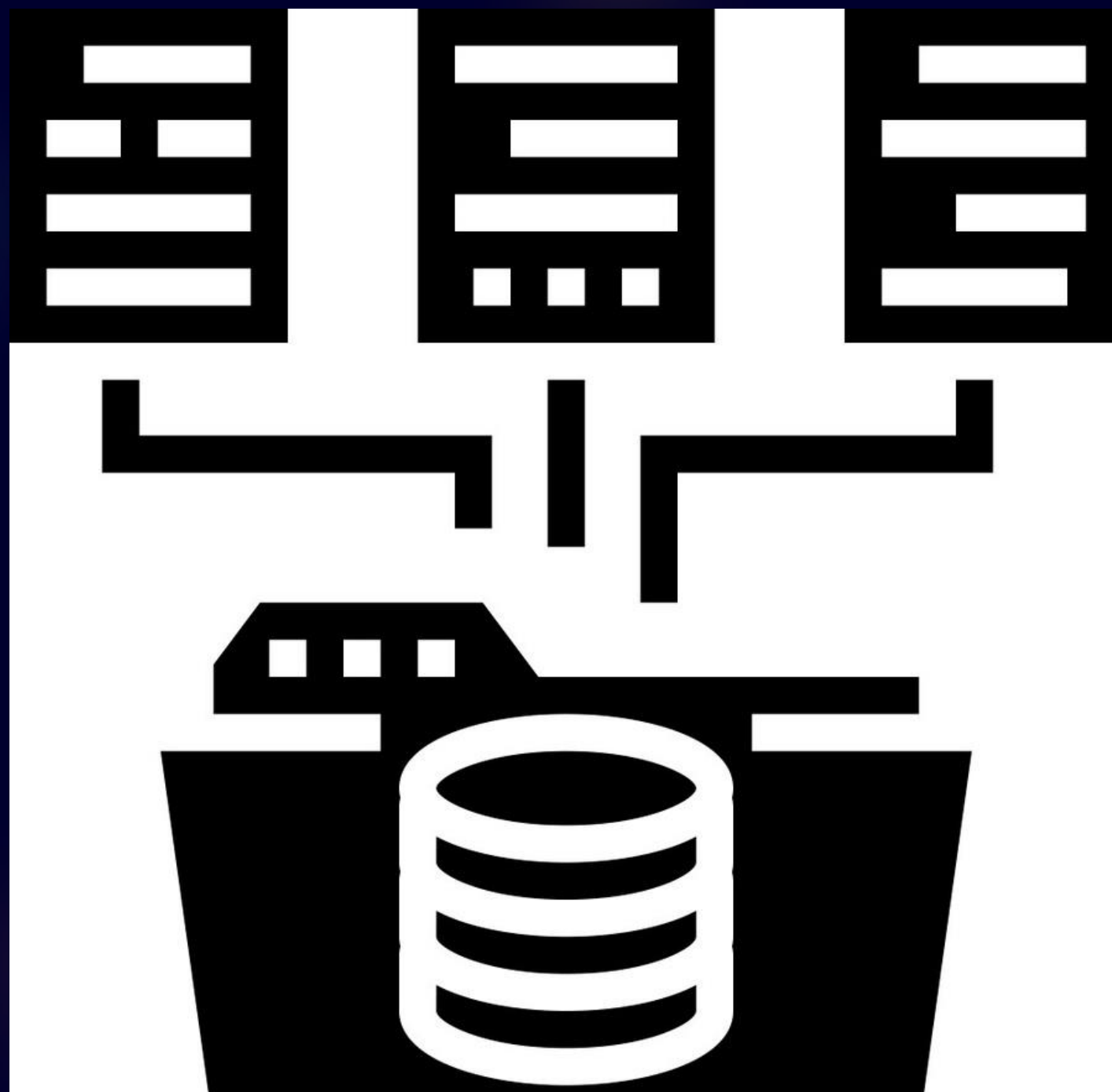
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DOISR-Core

Películas, actores, directores y productores con enfoque de género

IMDb Ratings

Puntuación y número de votos

Oscar Nominees (1929–present)

Nominaciones y ganadores por categoría y ceremonia

DOISR-Core: Estructura de datos

Dataset principal con cuatro entidades



Directores

ID, nombre, género, primera/última película, recuento de pronombres y créditos de actores/actrices.



Productores

ID, nombre, género, películas, recuento de pronombres y créditos.



Actores

ID, nombre, género y lista de películas.



Películas

ID, título, año, género, presupuesto ajustado, actores, directores y productores.

Requisitos de Competencia



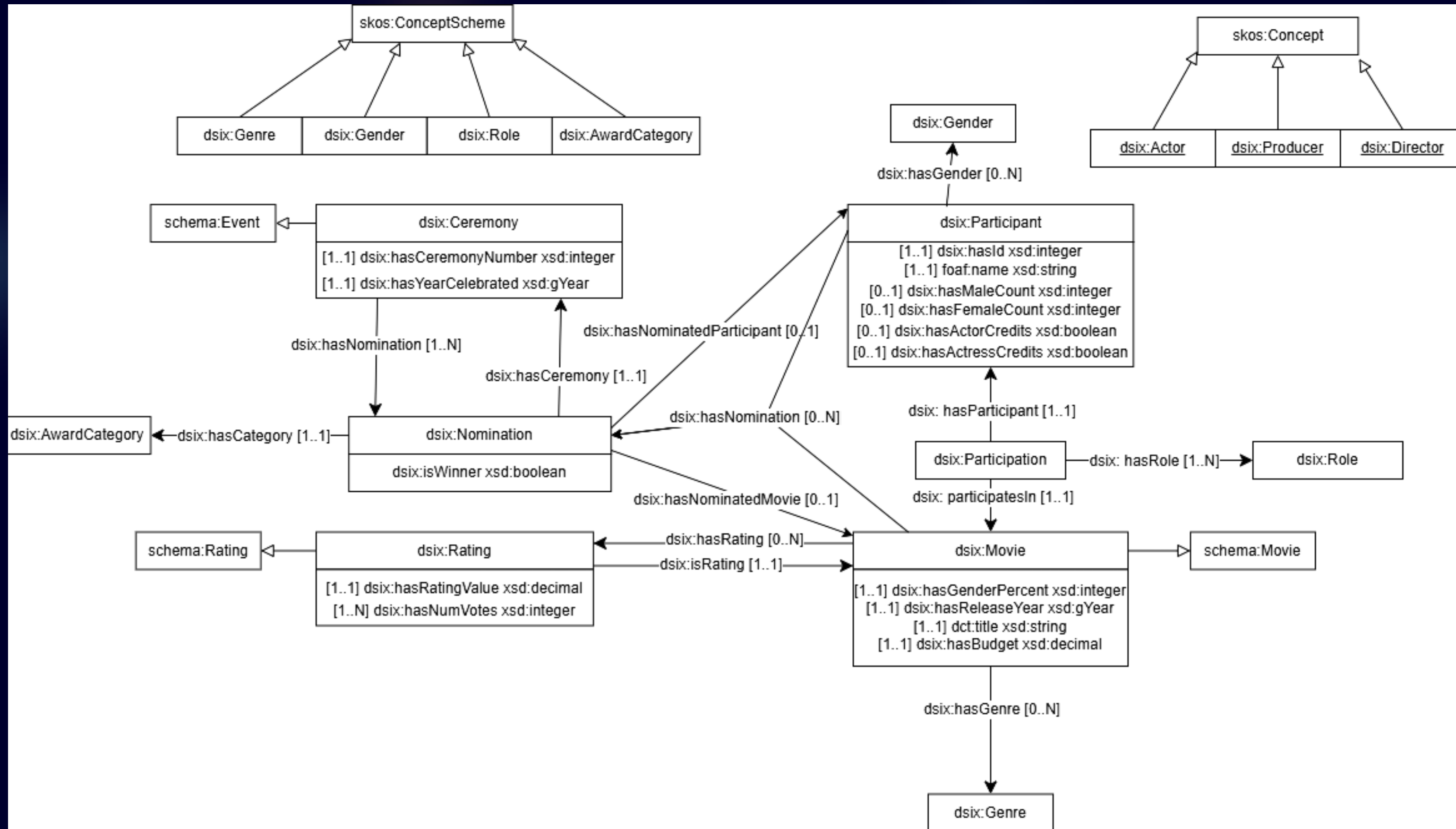
43 requisitos totales

- **Core R1-R33**
- **Ratings R34-R36**
- **Awards R37-R43**

Ejemplos destacados:

- R6 - Which actors participated in a given movie?
- R9 - What gender percentage has a given movie?
- R35 - What is the average rating of a given movie?
- R39 - What movies have been nominated for Academy Awards?

Conceptualización CHOWLK



Recursos Ontológicos Reutilizados



RDF / RDFS / OWL

Definición de clases, propiedades y restricciones.



Schema.org

Clases como *Movie*, *Rating* y *Event*.



FOAF

Nombres de participantes.



Dublin Core / SKOS / XSD

Metadatos y títulos de películas, vocabularios controlados y tipos de datos.

```
@base <https://w3id.org/doisr/Group6#> .
```

```
@prefix dsix: <https://w3id.org/doisr/Group6#> .
```

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
```

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
```

```
@prefix owl: <http://www.w3.org/2002/07/owl#> .
```

```
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

```
@prefix dct: <http://purl.org/dc/terms/> .
```

```
@prefix vann: <http://purl.org/vocab/vann/> .
```

```
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
```

```
@prefix schema: <https://schema.org/> .
```

```
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
```

```
<https://w3id.org/doisr/Group6> rdf:type owl:Ontology ;  
vann:preferredNamespaceUri "https://w3id.org/doisr/Group6#" ;  
vann:preferredNamespacePrefix "dsix" ;
```

```
# Basic ontology metadata
```

```
rdfs:label "IMDb Movie Ontology"@en ;
```

```
dct:title "IMDb Movie Ontology"@en ;
```

```
dct:description "Ontology for representing movies, actors, directors, and Oscar  
nominations."@en ;
```

```
owl:versionIRI <https://w3id.org/doisr/Group6/2.0.0> ;
```

```
owl:versionInfo "2.0.0" ;
```

```
dct:created "2026-04-20"^^xsd:date ;
```

```
dct:modified "2026-05-7"^^xsd:date ;
```

```
dct:creator "DOISR Group 6" ;
```

```
dct:contributor "Daniel Martín" ;
```

```
dct:contributor "Daniel Jimenez" ;
```

```
dct:contributor "Alejandro de Hermenegildo" ;
```

```
dct:contributor "Santiago Rodríguez" ;
```

```
dct:license <https://creativecommons.org/licenses/by/4.0/> ;
```

```
schema:codeRepository <https://github.com/LordDaniel-dev/DOISR-Group-6> ;
```

```
rdfs:seeAlso <https://www.imdb.com/> .
```

Implementación de la ontología

Implementación

- Paso de conceptualización en **diagrams.net** a **OWL** con **Chowlk Converter**
- Revisión y corrección de la ontología en **Protégé**
- Exportación de la ontología a **.ttl** o **.rdf**



Evaluación de la ontología

Evaluación

- Uso de la herramienta **OOPS!:**
4 pitfalls detectados inicialmente (1 importante)
- Corrección de P10 (axiomas disjuntos) y P13 (una parte)
- Pitfalls menores pendientes: P04, P08 (rdfs:comment) y P13

Pitfalls detected:		
Results for P04: Creating unconnected ontology elements.	5 cases	Minor
Results for P08: Missing annotations.	43 cases	Minor
Results for P13: Inverse relationships not explicitly declared.	7 cases	Minor

Results for P04: Creating unconnected ontology elements.5 casesMinor

Ontology elements (classes, object properties and datatype properties) are created isolated, with no relation to the rest of the ontology.

- This pitfall appears in the following elements:
 - > <http://www.w3.org/2004/02/skos/core#ConceptScheme>
 - > <https://schema.org/Movie>
 - > <https://schema.org/Rating>
 - > <http://www.w3.org/2004/02/skos/core#Concept>
 - > <https://schema.org/Event>

Results for P13: Inverse relationships not explicitly declared.7 casesMinor

This pitfall appears when any relationship (except for those that are defined as symmetric properties using owl:SymmetricProperty) does not have an inverse relationship (owl:inverseOf) defined within the ontology.

- This pitfall appears in the following elements:
 - > <https://w3id.org/doisr/Group6#hasCategory>
 - > <https://w3id.org/doisr/Group6#participatesIn>
 - > <https://w3id.org/doisr/Group6#hasNominatedMovie>
 - > <https://w3id.org/doisr/Group6#hasParticipant>
 - > <https://w3id.org/doisr/Group6#hasGenre>
 - > <https://w3id.org/doisr/Group6#hasRole>
 - > <https://w3id.org/doisr/Group6#hasGender>

Nivel de FAIRness

Uso de la herramienta **FOOPS!** para evaluar y mejorar el nivel de FAIRness de la ontología.

Findable

Identificadores únicos y metadatos ricos (F1–F3).

Accessible

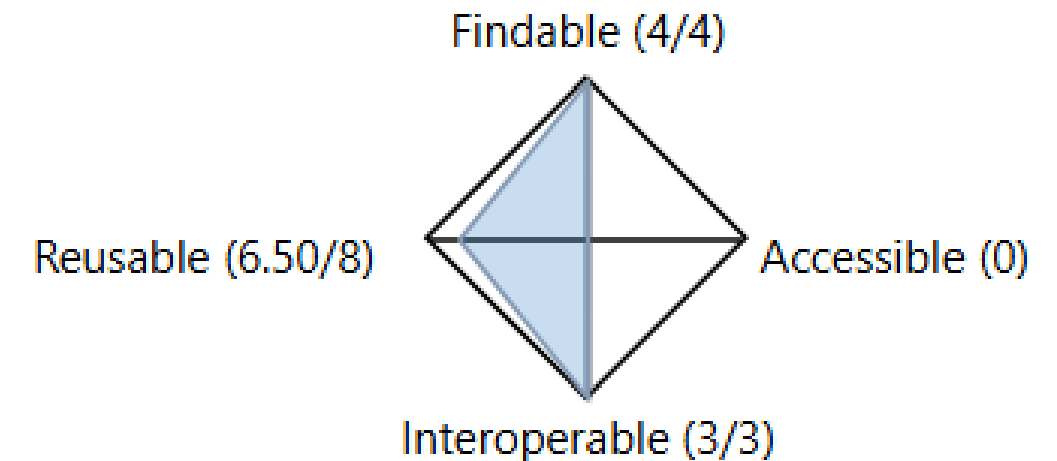
No evaluado por falta de URI al inicio.

Interoperable

Lenguaje formal y vocabularios FAIR (I1–I2).

Reusable

Pitfalls pendientes (R1): OM3 (50%) y VOC4.



Nivel de FAIRness (II)

Pitfalls restantes de Reusability:

OM3: Ontology declares detailed metadata

50%

^

Description: This test verifies if the ontology includes the following detailed metadata:

- Digital Object Identifier (DOI): declared using [bibo:doi](#), [schema:identifier](#) (if a doi is provided) or [dcterms:identifier](#) (if a doi is provided)
- publisher: declared using [dc:publisher](#), [dcterms:publisher](#) or [schema:publisher](#)
- logo: declared using [foaf:logo](#) or [schema:logo](#)
- status: declared using [bibo:status](#) or [mod:status](#)
- source: declared using [dcterms:source](#) or [prov:hadOriginalSource](#)
- issued date: declared using [dcterms:issued](#)
- previous version (optional): declared using [dc:replaces](#), [dcterms:replaces](#), [prov:wasRevisionOf](#), [owl:priorVersion](#) or [pav:previousVersion](#)
- backward compatibility (optional): declared using [owl:backwardCompatibleWith](#)
- modified date (optional): declared using [dcterms:modified](#) or [schema:dateModified](#)

The test will pass if all the detailed metadata properties are available in the ontology metadata (using any of the vocabularies listed above). The test will also check if previous version, backward compatibility and modified date are present, but with no penalty (as not all ontologies have a previous version).

Explanation: The following metadata was not found: doi, logo, status. Warning: The following OPTIONAL detailed metadata could not be found: previous version, backwards compatibility. Please consider adding them if appropriate.

VOC4: Ontology documentation: all terms have definitions

0%

^

Description: This check verifies whether all ontology terms have descriptions. The test will pass if all classes, properties and data properties have at least one [rdfs:comment](#), [skos:definition](#) or [obo:IAO_0000118](#) annotation. For skos vocabularies, only the skos:Concepts are assessed.

The test will fail otherwise, showing the level of completeness obtained (i.e. the percentage of documented terms).

Explanation: Descriptions found for 0 out of 36 terms.

SHACL Shapes

Ejemplo:

```
dsix:ParticipantShape a sh:NodeShape ;  
  sh:targetClass dsix:Participant ;
```

```
  sh:property [  
    sh:path schema:name ;  
    sh:datatype xsd:string ;  
    sh:minCount 1 ;  
    sh:maxCount 1 ;  
    sh:minLength 1 ;  
  ] ;
```

```
  sh:property [  
    sh:path dsix:hasGender ;  
    sh:in ( "female" "male" "other" ) ;  
    sh:minCount 1 ;  
    sh:maxCount 1 ;  
  ] .
```



Cambios realizados a SHAPES - Grupo 8

Mappings:

Clases:

- Movie <---->Movie
- Nomination <----> Nomination
- Ceremony <----> Ceremony
- Participant<---->Person

Relaciones:

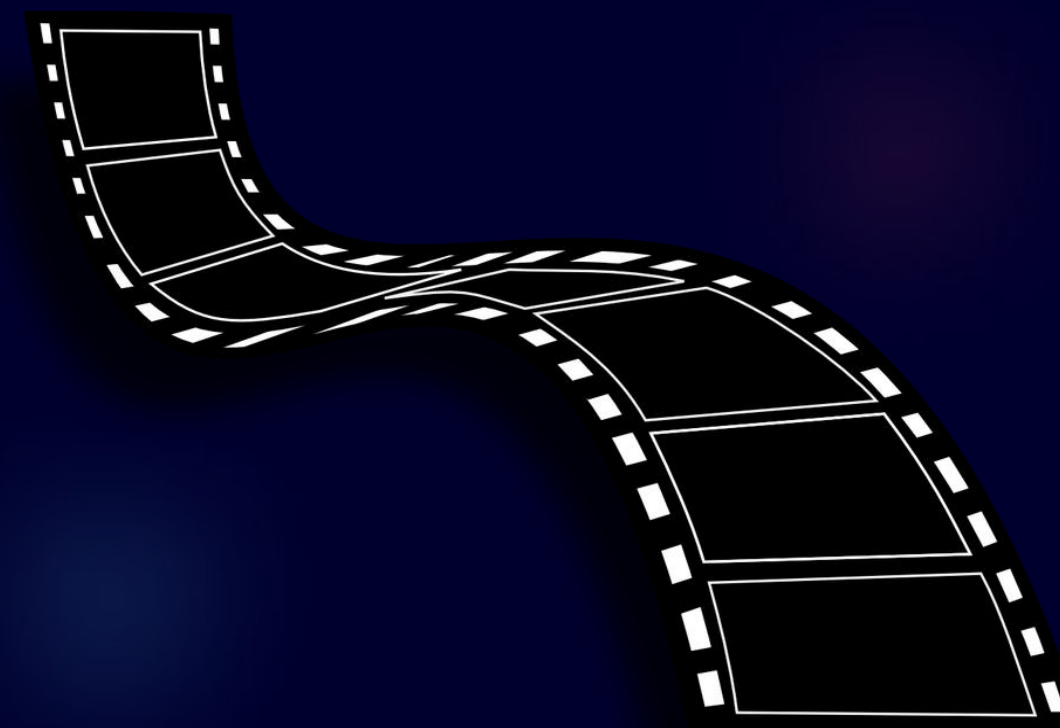
- isNominatedForMovie<---->hasNominatedMovie
- hasNominated<---->hasNominatedParticipant
- happenedAt<---->hasCeremony
- appearsIn<---->participatesIn

Atributos:

- nomineesIds<---->hasId
- hasLastMovieYear<---->hasLastMovieYear

Ejemplo de atributo añadido:

```
sh:property [  
  sh:path schema:dateOfBirth ;  
  sh:datatype xsd:date ;  
  sh:minCount 0 ;  
  sh:maxCount 1 ;  
];
```



Validación de Correcta Grupo 8

```
dsix:Oscars1928 a schema:Event ;  
dsix:year 1928 ; # Ahora es un integer válido >= 1927 dsix:ceremonyNumber 1 .
```

```
dsix:CharlesChaplin a schema:Person ;  
schema:name "Charles Chaplin"^^xsd:string ;  
dbo:religion "Atheist" ;  
dbo:race "White" ;  
dbo:sexualOrientation "Straight" .
```

```
dsix:TheCircus a schema:Movie ; schema:name "The Circus"^^xsd:string .
```

```
dsix:CharacterTramp a dsix:Character ;  
schema:name "The Tramp"^^xsd:string .
```

```
dsix:Nomination1 a dsix:Nomination ;  
schema:nominee :CharlesChaplin ; dsix:isNominatedforMovie :TheCircus ;  
dsix:isWinner "true"^^xsd:boolean .
```

```
pyshacl -s shapesConjunto.ttl  
PruebaGrupo8Correcto.ttl
```

Validation Report

Conforms: True

Validación de Incorrecta Grupo 8

```
:BadNomination a dsix:Nomination ;  
  dsix:isNominatedforMovie "Titanic the movie" ;  
  dsix:isWinner "Sí, ganó"^^xsd:string .
```

Falla Shape 2 y 3: Año es menor a 1927 (falla minInclusive) y Ceremony es 0 (falla minInclusive).

```
:BadEvent a schema:Event ;  
  dsix:year 1920 ;  
  dsix:ceremonyNumber 0 .
```

Falla Shape 4 y 5: Nombre vacío y religión inventada.

```
:BadPerson a schema:Person ;  
  schema:name ""^^xsd:string ;  
  dbo:religion "Jedi" ;  
  dbo:race "White" ;  
  dbo:sexualOrientation "Straight" .
```

Falla Shape 6: Le metemos una propiedad inventada a una clase cerrada.

```
:BadCharacter a dsix:Character ;  
  schema:name "Batman"^^xsd:string ;  
  dsix:weapon "Batarang"^^xsd:string .
```

```
pyshacl -s shapesConjunto.ttl
```

PruebaGrupo8Incorrecto.ttl Validation Report
Conforms:

False Results (6):

Constraint Violation in ClosedConstraintComponent (<http://www.w3.org/ns/shacl#ClosedConstraintComponent>): Severity: sh:Violation Source Shape:

:CharacterShape Focus Node: :BadCharacter Value Node: Literal("Batarang", datatype=xsd:string) Result Path: :weapon Message: Node :BadCharacter is closed. It cannot have value: Literal("Batarang", datatype=xsd:string)

Publicación



Ontología publicada



<https://w3id.org/doisr/Group6>



JSON LD – RDF – TTL – NTriples

Ontology Specification Draft

Release 2026-05-07

language [en](#)

This version:
<https://w3id.org/doi/r/Group6/2.0.0>

Revision:
2.0.0

Authors:
DOISR Group 6

Contributors:
Alejandro de Hermenegildo
Daniel Jimenez
Daniel Martin
Santiago Rodriguez

Publisher:
DOISR Group 6

Download serialization:

[Format JSON LD](#)
[Format RDF/XML](#)
[Format N Triples](#)
[Format TTL](#)

License: <https://creativecommons.org/licenses/by/4.0/>

Cite as:
Grupo 6. Ontología de cine. 2026

[Provenance of this page](#)

Abstract

Ontology for representing movies, actors, directors, and Oscar nominations.

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 - 1.1. [Namespace declarations](#)
- 2. [IMDb Movie Ontology: Overview](#)
- 3. [IMDb Movie Ontology: Description](#)
- 4. [Cross reference for IMDb Movie Ontology classes, properties and datatypes](#)
 - 4.1. [Classes](#)
 - 4.2. [Object Properties](#)
 - 4.3. [Data Properties](#)
 - 4.4. [Named Individuals](#)
- 5. [References](#)
- 6. [Acknowledgements](#)

1. Introduction

This is a place holder text for the introduction. The introduction should briefly describe the ontology, its motivation, state of the art and goals.

1.1. Namespace declarations

Table 1: Namespaces used in the document

dsix	< https://w3id.org/doisr/Group6 >
schema	< https://schema.org >
u-s-movies-with-gender-disambiguated-actors-directors-and-producers	< https://figshare.com/articles/dataset/U_S_movies_with_gender_disambiguated_actors_directors_and_producers >
mod	< https://w3id.org/mod >
owl	< http://www.w3.org/2002/07/owl >
doisr	< https://w3id.org/doisr >
xsd	< http://www.w3.org/2001/XMLSchema >
skos	< http://www.w3.org/2004/02/skos/core >
rdfs	< http://www.w3.org/2000/01/rdf-schema >
0-1	< http://xmllns.com/foaf/0.1 >
rdf	< http://www.w3.org/1999/02/22-rdf-syntax-ns >
terms	< http://purl.org/dc/terms >
vann	< http://purl.org/vocab/vann >
lorddaniel-dev	< https://github.com/LordDaniel-dev >

2. IMDb Movie Ontology: Overview

This ontology has the following classes and properties.

Classes

Award Category	Ceremony	Gender	Genre	Movie	Nomination	Participant	Participation	Rating	Role
--------------------------------	--------------------------	------------------------	-----------------------	-----------------------	----------------------------	-----------------------------	-------------------------------	------------------------	----------------------

Object Properties

[has category](#) [has ceremony](#) [has gender](#) [has genre](#) [has nominated movie](#) [has nominated participant](#) [has nomination](#)
[has nomination participant](#) [has participant](#) [has rating](#) [has role](#) [is rating](#) [participates in](#)

```

JSON Datos sin procesar Cabeceras
Guardar Copiar Contraer todo Expandir todo Filtar JSON

▼ 0:
  @id: "_" :genidi1"
  ▼ @type:
    0: "http://www.w3.org/2002/07/owl#Restriction"
  ▼ http://www.w3.org/2002/07/owl#minQualifiedCardinality:
    ▼ 0:
      @type: "http://www.w3.org/2001/XMLSchema#nonNegativeInteger"
      @value: "1"
  ▼ http://www.w3.org/2002/07/owl#onClass:
    ▼ 0:
      @id: "https://w3id.org/doisr/Group6#Nomination"
  ▼ http://www.w3.org/2002/07/owl#onProperty:
    ▼ 0:
      @id: "https://w3id.org/doisr/Group6#hasNomination"
▼ 1:
  @id: "_" :genidi0"
  ▼ @type:
    0: "http://www.w3.org/2002/07/owl#Restriction"
  ▼ http://www.w3.org/2002/07/owl#maxQualifiedCardinality:
    ▼ 0:

@prefix : <https://w3id.org/doisr/Group6> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix xml: <http://www.w3.org/XML/1998/namespace> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@base <https://w3id.org/doisr/Group6> .

<https://w3id.org/doisr/Group6> rdf:type owl:Ontology ;
    owl:versionIRI <https://w3id.org/doisr/Group6/2.0.0/> ;
    <http://purl.org/dc/terms/bibliographicCitation> "Grupo 6. Ontología de cine. 2026." ;
    <http://purl.org/dc/terms/contributor> "Alejandro de Hermenegildo" ,
        "Daniel Jimenez" ,
        "Daniel Martin" ,
        "Santiago Rodriguez" ;
    <http://purl.org/dc/terms/created> "2026-04-20"^^xsd:date ;
    <http://purl.org/dc/terms/creator> "DOISR Group 6" ;
    <http://purl.org/dc/terms/description> "Ontology for representing movies, actors, directors, and Oscar nominations."@en ;
    <http://purl.org/dc/terms/issued> "2026-05-18"^^xsd:date ;
    <http://purl.org/dc/terms/license> <https://creativecommons.org/licenses/by/4.0/> ;
    <http://purl.org/dc/terms/modified> "2026-05-07"^^xsd:date ;
    <http://purl.org/dc/terms/publisher> "DOISR Group 6" ;
    <http://purl.org/dc/terms/source> <https://figshare.com/articles/dataset/U_5_movies_with_gender-disambiguated_actors_directors_and_producers/4967876> ;
    <http://purl.org/vocab/vann/preferredNamespacePrefix> "dsix" ;
    <http://purl.org/vocab/vann/preferredNamespaceUri> <https://w3id.org/doisr/Group6#> ;
    rdfs:label "IMDb Movie Ontology"@en ;
    rdfs:seeAlso <https://www.imdb.com/> ;
    owl:versionInfo "2.0.0" ;
    <https://schema.org/codeRepository> <https://github.com/LordDaniel-dev/DOISR-Group-6> ;
    <https://w3id.org/modecreatedwith> <https://chowik.linkeddata.es/> .

#####
# Annotation properties
#####

## http://purl.org/dc/terms/bibliographicCitation
<http://purl.org/dc/terms/bibliographicCitation> rdf:type owl:AnnotationProperty .

## http://purl.org/dc/terms/contributor
<http://purl.org/dc/terms/contributor> rdf:type owl:AnnotationProperty .

## http://purl.org/dc/terms/created

-<rdf:RDF xml:base="https://w3id.org/doisr/Group6">
  -<owl:Ontology rdf:about="https://w3id.org/doisr/Group6">
    <owl:versionIRI rdf:resource="https://w3id.org/doisr/Group6/2.0.0/">
    <terms:bibliographicCitation>Grupo 6. Ontología de cine. 2026.</terms:bibliographicCitation>
    <terms:contributor>Alejandro de Hermenegildo</terms:contributor>
    <terms:contributor>Daniel Jimenez</terms:contributor>
    <terms:contributor>Daniel Martin</terms:contributor>
    <terms:contributor>Santiago Rodriguez</terms:contributor>
    <terms:created rdf:datatype="http://www.w3.org/2001/XMLSchema#date">2026-04-20</terms:created>
    <terms:creator>DOISR Group 6</terms:creator>
  -<terms:description xml:lang="en">
    Ontology for representing movies, actors, directors, and Oscar nominations.
  </terms:description>
    <terms:issued rdf:datatype="http://www.w3.org/2001/XMLSchema#date">2026-05-18</terms:issued>
    <terms:license rdf:resource="https://creativecommons.org/licenses/by/4.0/">
    <terms:modified rdf:datatype="http://www.w3.org/2001/XMLSchema#date">2026-05-07</terms:modified>
    <terms:publisher>DOISR Group 6</terms:publisher>
    <terms:source rdf:resource="https://figshare.com/articles/dataset/U_5_movies_with_gender-disambiguated_actors_directors_and_producers/4967876/">
    <vann:preferredNamespacePrefix>dsix</vann:preferredNamespacePrefix>
    <vann:preferredNamespaceUri>https://w3id.org/doisr/Group6#</vann:preferredNamespaceUri>
    <rdfs:label xml:lang="en">IMDb Movie Ontology</rdfs:label>
    <rdfs:seeAlso rdf:resource="https://www.imdb.com/">
    <owl:versionInfo>2.0.0</owl:versionInfo>
    <schema:codeRepository rdf:resource="https://github.com/LordDaniel-dev/DOISR-Group-6">
    <mod:createdWith rdf:resource="https://chowik.linkeddata.es/">
  </owl:Ontology>
-</--

#####
//
// Annotation properties
//
#####

-->

```

Uso de IA



Búsqueda de información

Dar feedback

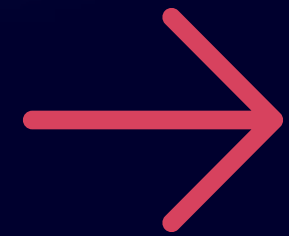
Interpretar errores

Diseño de plantillas

Conclusiones



Ontología completa



Objetivos FAIR



Integración con otras ontologías



Gracias por vuestra atención

Grupo 6

Developing Open Interoperable Semantic Resources